MILITARY IS RELYING ON



explosive warhead

Role: Used for short-range precision strikes

TAPAS-BH-201
Medium-Altitude Long-Endurance (MALE) UAV developed by DRDO

Role: Designed for ISR (intelligence, surveillance, reconnaissance) missions



Arriving Soon

PROJECT KAL
Being designed by IG Defence as 'deep strike' platform to be used in waves

Role: Ideal for precision strikes against infrastructure like command centres, fuel depots

SHESHNAG-150
Being designed by New-Space Research, this is an AI-driven loitering munition

Role: Equipped with a high-explosive warhead, it is ideal for saturating air defences

THE ARMY DESIGN BUREAU WAS TASKED WITH FORMING A FRAMEWORK TO ASSESS CYBERSECURITY THREATS IN MILITARY DRONES. IT IS PENDING APPROVAL WITH THE DEFENCE MINISTRY



A CERTIFICATION GAP

Such concerns are widely shared and the Army Design Bureau was tasked with developing a framework to assess and mitigate cybersecurity vulnerabilities in military drones, following consultations with the armed forces and the industry. However, the framework has reportedly remained pending with the ministry of defence for nearly 18 months.

Sai Pattabiram of Chennai-based Zuppa Drones—which has supplied the army with tactical UAVs)—points to the existing certification requirements for CCTV systems to argue that India should make STQC (Standardisation Testing and Quality Certification Directorate, under the Union ministry of electronics and information technology) certification mandatory for drones. "Drones and CCTV cameras share the same core electronics and the same cybersecurity risk—an undisclosed connection to a foreign server," he says. "CCTV certification is mandatory, yet under the Airworthiness and Drone-Based framework, it remains voluntary for drones. That is a serious anomaly."

Drones, he argues, present a greater security challenge because they are mobile, capable of operating across borders and contested airspace, and can be weaponised. "A compromised drone can expose the operator's location, ISR data and command systems, or even allow hijacking," Pattabiram says. "This has implications for national security, defence readiness and India's ambitions as a global drone-manufacturing hub." He says that testing should include mandatory disclosure of server connections and the ability to conduct teardown inspections of critical components. "India has already developed this capability," says Pattabiram. "The question is why it has not yet been made mandatory."

Maintaining visibility deep into complex defence supply chains is a global concern. A 2025 US Government Accountability Office investigation identified Chinese-origin components in the F-35 fighter jet supply chain and Chinese involvement in parts of the MQ-9 Reaper drone supply chain, despite the platforms being produced by major US or western firms. Supply-chain security is paramount as military drones become more deeply integrated with surveillance, communications and command networks.

India, therefore, needs a mandatory and independent security-verification regime for military UAVs—one that traces components deep into the supply chain and tests how they behave in the real world. ■